

W6 - The Ambiguous Case of Sine

MCR3U

1) In $\triangle ABC$, $a = 13$ cm, $b = 21$ cm, and $\angle A = 29^\circ$. Draw possible diagrams that match the given measurements. Then calculate the length of side c .

2) In $\triangle ABC$, $a = 5.9$ m, $b = 7.8$ m, and $\angle A = 36^\circ$. Draw possible diagrams that match the given measurements. Then calculate the length of side c .

3) In $\triangle ABC$, $a = 2.4$ cm, $c = 3.2$ cm, and $\angle A = 28^\circ$. Determine two possible measures for $\angle C$ and for the length of side b .

4) In $\triangle DEF$, $d = 3$ cm, $e = 5$ cm, and $\angle D = 30^\circ$. Determine two possible measures for $\angle E$ and for the length of side f .

5) Two ships, S and T, are 120 km apart when they pick up a distress call from a yacht. Ship T estimates that the yacht is 70 km away and that the angle between the line from T to S and the line from S to the yacht is 28° . What are two possible distances, to the nearest tenth of a km, from ship S to the yacht?

Answers

- 1) 26.5 cm or 10.3 cm
- 2) 10 m or 2.6 m
- 3) $\angle C = 39^\circ$ and $b = 4.7$ cm; $\angle C = 141^\circ$ and $b = 1.0$ cm
- 4) $\angle E = 56^\circ$ and $f = 6$ cm; $\angle E = 124^\circ$ and $f = 2.7$ cm
- 5) 147.5 km or 64.4 km